LCGC RFQ & NPP Procurement Automation Platform

Complete System Flow — Login to Purchase Order

1. Login, Signup & User Management

1

Signup / Login

Users can create an account and log in either with email-password or directly with Google Sign-In (Login with Google) for faster, authenticated single sign-on access.

2

User & Role Creation

Admin creates user IDs and assigns dedicated roles to each user — Admin, Engineer, Purchase Head, VP, Vendor, Finance, and other corporate designations as required.

3

Role-Based Access Control (RBAC)

Every screen and action is permission-controlled: a user can only view or perform actions their assigned role has been given rights for. If a role is not granted access to a module, that user cannot open or act on it — ensuring only authorised people (e.g. only Purchase Head/VP for approvals) can perform sensitive actions.

- Strict authorization gates prevent unauthorized viewing of competitor pricing.
- Granular permissions for Create, Review, Approve, Reject, and PO Release.

2. RFQ Creation & Vendor Quotation

1

Create RFQ

An authorised user (Engineer/Requisitioner) creates a new RFQ with item details, specifications, quantity, target price, and required delivery date, then selects which vendors to send it to.

2

Submit & Auto-Send to Vendors

On submission, the RFQ (requisition) is automatically sent to all the selected vendors — no manual emailing needed. Vendors receive instantaneous notification with direct access link.

3

Vendor Accepts & Responds

Each invited vendor receives the RFQ, accepts it in their portal, and submits their quotation — attaching the unit price, GST rate, and HSN code, along with supporting documents (quotes, certificates, catalogs).

4

Multiple Vendor Quotations

Since the RFQ goes to multiple vendors, each vendor submits their own separate quotation independently, so the system collects several competing offers for the same requirement.

3. Cost Comparison & Recommendation

1

Automated Cost Comparison

Once vendor quotations are in, the system automatically compares all the vendors' quotations side by side on price, GST, delivery time, and other commercial terms without manual spreadsheet compilation.

- Computes Landed Price = Basic Unit Rate + Packaging + Freight + Applicable GST.
- Identifies tax differences, HSN discrepancies, and payment terms across suppliers.

2

Best Vendor Recommendation (L1 / L2 / L3 Ranking)

Based on the comparison, the system recommends the best (lowest-cost/most suitable) vendor for the requirement — automatically generating L1, L2, and L3 ranking to pick the winning bid transparently.

- Clear commercial justification logged automatically for procurement audits.
- Procurement officers can record negotiation rounds and final agreed price.

4. Purchase Request (PR)

1

PR Creation (Zero Re-entry)

Once a vendor is selected from the cost comparison, a Purchase Request is generated/raised with the recommended vendor's details, item specifications, unit price, GST, and terms 100% auto-filled.

- Eliminates manual re-typing and transcription errors from quotation to PR.
- Includes attached drawings, vendor technical catalog, and comparison sheet.

2

PR Approval Steps & Strict Hierarchy

The Purchase Request goes through its own defined approval steps/chain (role-based, e.g. Purchase Head, VP, etc.) before it can move ahead to Purchase Order generation.

- Strict sign-off rule: All approvers in the chain must sign off before PR becomes Approved.
- Status remains "In Process" while any approver is pending, eliminating accidental approvals.

3

Real-time Days Pending & SLA Tracking

Every active PR displays a live indicator showing how many days it has been waiting (e.g. #ó 3 Days Pending). Bottlenecks are highlighted immediately for executive intervention.

5. EP (Early Payment / Expense) Request

1

Create EP Request Form

An EP Request Form can be created/filled for early payment or expense-related requests linked to the purchase, complete with title, department, amount, objective, and priority level.

- Supports attaching invoices, debit notes, proforma invoices, and vendor receipts.
- Auto-populates requester details, contact number, and organization profile.

2

Approval Request Status & Segregation

The EP request goes through its own approval chain, and its live status (Pending / In Process / Approved / Rejected) is tracked and visible as 'Approval Request Status'.

- Parallel & Sequential Workflow: Supports both parallel approvers and sequential manager lines.
- Auto-Archive: Approved and Rejected requests automatically move out of active requests into Status tab.

3

In-App PDF Preview, Download & Official Print

Approvers and auditors can preview the official EP document directly in the browser, download high-definition PDF reports, or print receipt-style hard copies with complete sign-off history.

4

Direct Fast Email Sharing

Users can enter any recipient email address in the View modal to instantly send the complete approved request data along with the official PDF attachment in seconds.

6. Purchase Order (PO) Generation

1

PO Generation (Formal Receipt-Style)

After the required approvals (PR/EP) are completed, the Purchase Order is generated automatically — a formal, receipt-style document with all item, vendor, price, GST/HSN, and terms prefilled from the approved PR.

- Generates unique Purchase Order Number (e.g. PO-NPP-2026-XXXX).
- Includes billing address, shipping destination, delivery timeline, and payment terms.

2

Dispatch to Vendor & Digital Acceptance

The finalised PO is shared with the selected vendor via automated email and published on their portal. The vendor logs in, confirms acceptance, and proceeds with manufacturing/delivery.

- Real-time vendor acceptance timestamp and delivery milestone tracking.
- Seamless handoff to Stores for Goods Receipt Note (GRN) and Quality inspection (DQMS).

THE 4 CORE PROCUREMENT PHASES: ARCHITECTURAL BREAKDOWN

PHASE 1
STAGE

Demand Raising & RFQ Floatation

- Material Requirement: Requisitioner initiates item demand with part code, specifications, target date, and estimated budget.
- Document Attachments: Engineering drawings, specification sheets, and Excel BOMs uploaded and linked to tender.
- Departmental Validation: Section head verifies technical requirement before procurement involvement.
- Tender Publication: Procurement selects verified suppliers and floats RFQ with tender closing deadline.

PHASE 2
STAGE

Vendor Response, Bidding & SheetJS Viewer

- Supplier Portal: Vendors authenticate securely to view item specs and download CAD/technical drawings.
- Commercial Quotation: Direct entry of Basic Price, GST rate, HSN Code, delivery timeline, and payment terms.
- Interactive File Viewer: Integrated SheetJS & PDF engine allows instant preview of vendor Excel spreadsheets & catalogs.
- Multi-Round Negotiation: Procurement sends clarification queries; vendors submit revised quotes with full audit trails.

PHASE 3
STAGE

Automated Cost Comparison & Approval Matrix

- Comparative Matrix (CS): Auto-ranks participating vendors into L1, L2, L3 on landed cost without manual calculation.
- Auto-Prefilled PR: Winning vendor details, rates, and specifications transfer into Purchase Requisition with 0 re-typing.
- Sequential & Parallel Approvals: Supports multi-tier routing (Engineer -> Purchase Head -> VP-Operations -> Director).
- Strict Approval Rule: Request remains "In Process" until 100% of approvers in the chain approve.

PHASE 4
STAGE

Purchase Order Release, Delivery & Payment

- 1-Click PO Creation: Converts approved PR into formal legal Purchase Order with all commercial terms prefilled.
- Vendor Dispatch: Official branded PO PDF dispatched via automated email and posted to vendor dashboard.
- Delivery & GRN Tracking: Material receipt note (GRN) created at factory gate entry with defect logging in DQMS.
- 3-Way Matching: Automated verification of PO vs GRN vs Vendor Invoice for zero-error finance payment release.

End-to-End Flow Summary

COMPLETE STEP-BY-STEP WORKFLOW LIFECYCLE:

Login/Signup (incl. Google) !' Role-based User Access (Admin / Engineer / Purchase Head / VP / Vendor) !' Create RFQ !' RFQ auto-sent to Selected Vendors !' Vendor Accepts & Submits Quotation (Price, GST, HSN) from Multiple Vendors !' Automated Cost Comparison & Best Vendor Recommendation (L1/L2/L3) !' Purchase Request with Approval Steps !' EP Request Form with Approval Status Tracking (Pending / In Process / Approved) !' Purchase Order Generated (Receipt-style) & Dispatched to Vendor !' Goods Receipt & Payment Reconciliation.

7. Key Business Benefits & Measurable Impact (ROI)

65% Reduction in Cycle Time

PROCUREMENT TAT

Manual paper forms, follow-up phone calls, and email chasing eliminated completely.

8% - 14% Direct Savings

COST OPTIMIZATION

Transparent multi-vendor bidding and automatic L1 comparative evaluation reduce purchase costs.

100% Audit Compliance

GOVERNANCE & SECURITY

Every action logged with user ID, role, approval remarks, timestamp, and IP traceability.

Zero Transcription Errors

DATA INTEGRITY

100% prefilled data flow: RFQ details flow seamlessly to Quotation, PR, and Purchase Order.

PROJECT ARCHITECTURE & CLIENT SIGN-OFF:

- System: LCGC RFQ & NPP Procurement Automation Platform | Version: 2.5 (Enterprise Edition)
- Prepared for: Radiant Appliances & Electronics / Client Executive Review
- Verification Status: Fully tested, live end-to-end integration across all 4 procurement phases.